

Yet interest in flight had been stimulated afresh – not least in the heart of Sir George Cayley, who now embarked on a new round of experiments that culminated in the world's first manned heavier-than-air flight in a glider in 1853. The “pilot” was Cayley's coachman. He was reluctantly persuaded to climb into the boatlike fuselage of a glider, which then rolled down one side of a valley, lifted into the air, and briefly flew before coming down uncomfortably. The coachman is said to have immediately put in his notice, on the grounds that he had been “hired to drive, not to fly.” Cayley's success, achieved in the privacy of his estate, had less public impact than Henson's failure. The story of the coachman's flight only came to light long after Cayley's death in 1857.

High society

The growing respectability of flight research was exemplified by the foundation of the Aeronautical Society of Great Britain in 1866, a dignified association of scientists and engineers who staged the world's first exhibition of flying machines at London's Crystal Palace two years later. None of them flew. However, there were some notable efforts to advance understanding of aerodynamics.

Francis Wenham, a distinguished marine engineer and a founding member of the Aeronautical Society, built the first wind tunnel and produced improved data on the lift provided by different wing shapes. In the 1870s, brilliant young French engineer Alphonse Pénaud

“Give us a motor and we will very soon give you a successful flying machine.”

HIRAM MAXIM, 1892

made significant progress in wing and tail design through experiments with model aircraft powered by a twisted rubber band. And Louis-Pierre Mouillard's book *The Empire of the Air*, published in 1881, was an inspirational work based on the

author's observations of bird flight.

Practical efforts to progress in heavier-than-air manned flight in the late 19th century divided into three approaches. One focused on power: was it possible to find an engine powerful enough, or rather with a favorable enough power-to-weight ratio, to lift a machine and a man into the air? A second focused on unpowered flight as a means of

understanding the secret of flight as exhibited by birds. A third, like Pénaud, focused on model building. Success would only be achieved when the traditions of powered and unpowered flight came together – in the Wright brothers.

Engine power

Early experimenters in powered flight were unfortunate in that their only feasible power plant was a steam engine. The first of the steam-powered experimenters to make a serious attempt to get off the ground was a French naval officer, Félix du Temple de la Croix. In the 1850s, with his brother Louis, he designed and flew a model airplane powered first by clockwork and then by a miniature steam engine. He then patented a design for a full-size monoplane with a lightweight steam engine and the surprising refinement of a retractable undercarriage. His man-carrying airplane was finally built and ready to test in 1874. With a French sailor on board, it

ran down a sloping ramp, briefly lifted into the air, and immediately came back down to earth.

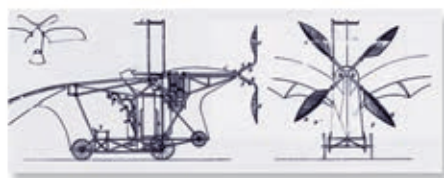
A decade later, du Temple's “hop” was matched by a Russian experimenter, Aleksander Mozhaikii. In 1884, at Krasnoe Selo outside St. Petersburg, Mozhaikii tested a two-engined monoplane with a mechanic at the helm. Spouting smoke from its ship-like funnel it momentarily lifted, then crashed to the ground.

The first claim to have actually cracked the problem of powered flight came from French electrical engineer Clément Ader. After testing his bat-winged, steam-powered *Éole* in 1890, he claimed: “I have resolved the problem after much work, fatigue, and money.” What he had achieved, as far as can be



FRENCH CANDIDATE

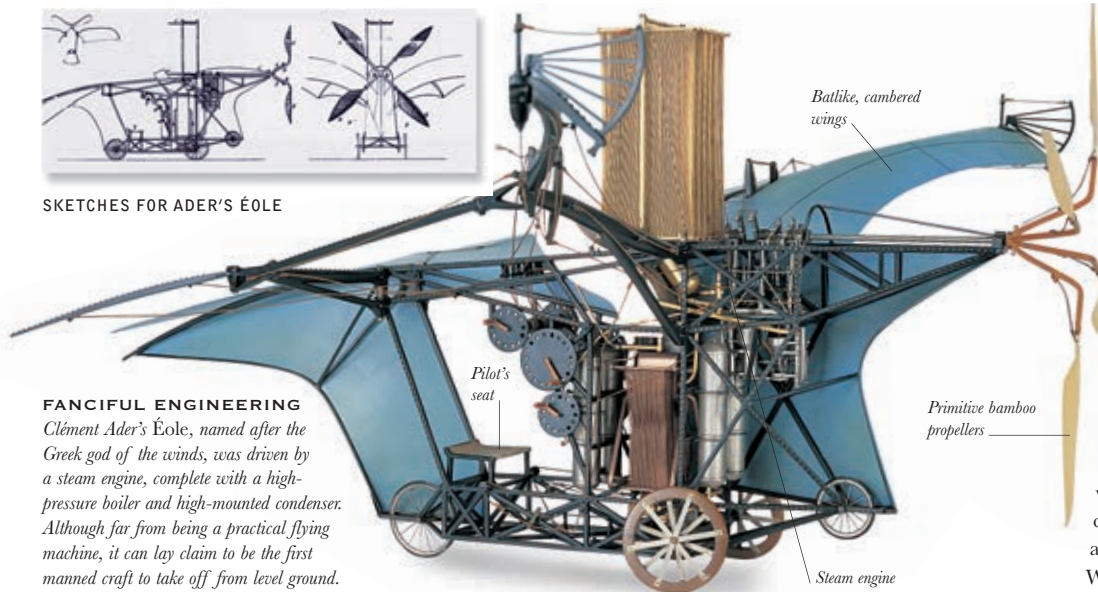
In the 1890s French electrical engineer Clément Ader built two steam-powered flying machines with wings modeled on a bat. Although his first, the *Éole* (below) managed to hop for 163ft (50m) the *Avion III*, seen above, failed to get off the ground.



SKETCHES FOR ADER'S ÉOLE

FANCIFUL ENGINEERING

Clément Ader's *Éole*, named after the Greek god of the winds, was driven by a steam engine, complete with a high-pressure boiler and high-mounted condenser. Although far from being a practical flying machine, it can lay claim to be the first manned craft to take off from level ground.



Batlike, cambered wings

Pilot's seat

Primitive bamboo propellers

Steam engine